SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code *of Conduct:*

I T.Sanhith(192311228) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

> **a. Compute n ϕ(n)**
> **b. Choose e=7 and find the private key d**
> **c. Encrypt the message M=9**
> **d. Decrypt the ciphertext and verify the result**
> **e. Analyze how changing e affects security**

ANS:
RSA Encryption Analysis
Given

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute $n$and $\phi(n)$
Step 1: Calculate n

$$n = p \times q$$
$$n = 11 \times 13 = 143$$

Step 2: Calculate Euler's Totient

$$\phi(n) = (p - 1)(q - 1)$$
$$= (11 - 1)(13 - 1)$$
$$= 10 \times 12 = 120$$

Answer

$$\boxed{n = 143,\ \phi(n) = 120}$$

(b) Find the private key d
We need

$$d \times e \equiv 1 \pmod{120}$$

Since

$$e = 7$$

Find inverse of 7 modulo 120.
Checking:

$$7 \times 103 = 721$$
$$721 \div 120 = 6 \text{ remainder } 1$$

Hence

$$\boxed{d = 103}$$

Private key

$$(d, n) = (103, 143)$$

(c) Encrypt Message $M = 9$
Formula

$$C = M^e \pmod n$$
$$C = 9^7 \pmod{143}$$

Calculate

$$9^2 = 81$$
$$9^4 = 81^2 = 6561$$
$$6561 \bmod 143 = 126$$

Then

$$9^7 = 9^4 \times 9^2 \times 9$$
$$= 126 \times 81 \times 9$$

First

$$126 \times 81 = 10206$$
$$10206 \bmod 143 = 53$$

Then

$$53 \times 9 = 477$$
$$477 \bmod 143 = 48$$

Hence

$$\boxed{C = 48}$$

(d) Decrypt the Ciphertext
Formula

$$M = C^d \pmod n$$
$$M = 48^{103} \bmod 143$$

Using modular exponentiation,

$$48^{103} \bmod 143 = 9$$

Recovered message

$$\boxed{M = 9}$$

Verification:
Original message = Decrypted message
✓ Verified successfully.

(e) Effect of changing e

- Smaller e
  - Faster encryption
  - May introduce vulnerabilities if padding is weak
- Larger e
  - Slower encryption
  - Slightly higher computational cost

Common value:

$$\boxed{e = 65537}$$

2. Block Cipher Mode Evaluation

Given plaintext blocks:

P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000) b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

ANS :

Given

Plaintext blocks

P1 = 1010

P2 = 1010

P3 = 1111

P4 = 1010

Key

K = 1100

IV

0000

Assume encryption uses XOR.

(a) ECB Mode

Formula

$$\boldsymbol{C = P \oplus K}$$

Block 1

1010

1100

0110

Block 2

Same plaintext

0110

Block 3

1111

1100

0011

Block 4

0110
Ciphertext
ECB = [0110,0110,0011,0110]

CBC Mode
Formula

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

Initial
IV=0000

Block 1
1010 XOR 0000 =1010

1010 XOR1100=0110
C1=0110

Block 2
1010 XOR0110=1100

1100 XOR1100=0000
C2=0000

Block 3
1111 XOR0000=1111

1111 XOR1100=0011
C3=0011

Block 4
1010 XOR0011=1001

1001 XOR1100=0101
C4=0101
Final CBC
CBC=[0110,0000,0011,0101]

(b) Compare Outputs
ECB
0110 0110 0011 0110
CBC
0110 0000 0011 0101
Repeated plaintext in ECB produces repeated ciphertext.
CBC produces different ciphertext even for identical plaintext blocks.

(c) Which mode hides patterns better?
CBC Mode
Reason:

- **Every block depends on previous ciphertext.**

- **Same plaintext generates different ciphertext.**
- **Prevents pattern leakage.**

Hence,

$\boxed{\textbf{CBC is more secure than ECB.}}$

3. DES vs 3-DES vs Blowfish Performance Study

Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

ANS :

(i) Total Encryption Time

DES
$1000 \times 2 = 2000$ ms =2 seconds

3DES
$1000 \times 6 = 6000$ ms =6 seconds

Blowfish
$1000 \times 1 = 1000$ ms =1 second

(ii) Trade-offs

DES

- **Small key**
- **Fast**
- **Not secure today**

3DES

- **Strong security**
- **Very slow**

Blowfish

- **Good security**
- **Fast**
- **Flexible key size**

(iii) Recommendation

For real-time communication

$\boxed{\textbf{Blowfish}}$

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

ANS

Given

$$p = 23$$
$$g = 5$$

Private keys
A=6
B=15

(d) Compute Public Keys
Formula

$$A = g^a \bmod p$$
$$= 5^6 \bmod 23$$
$$5^2 = 25 \equiv 2$$
$$5^4 = 2^2 = 4$$
$$5^6 = 5^4 \times 5^2$$
$$4 \times 2 = 8$$

Public key A

$$\boxed{8}$$

Public key B

$$5^{15} \bmod 23$$

Using modular arithmetic,

$$\boxed{19}$$

(e) Shared Secret
For A

$$19^6 \bmod 23$$

Result

$$\boxed{2}$$

For B

$$8^{15} \bmod 23$$

Result

$\boxed{2}$

Shared key

$\boxed{2}$

(f) Man-in-the-Middle Attack
Attacker intercepts both public keys.
Instead of sending
A → B : 8
Attacker sends
5
Similarly,
Instead of
19
Attacker sends another fake value.
Now

- **A shares secret with attacker.**
- **B shares another secret with attacker.**

Attacker decrypts, reads, modifies, and re-encrypts every message.

(g) Prevention
Use authentication:

- **Digital Certificates**
- **RSA Signatures**
- **Digital Signatures**
- **Public Key Infrastructure (PKI)**

Best answer:

$\boxed{\textbf{Use Digital Signatures with Diffie-Hellman.}}$

5. ECC vs RSA Comparison
Given:

a. **RSA key size = 1024 bits**
b. **ECC key size = 160 bits (equivalent security)**
c. **Device processing capacity = limited (IoT device)**
d. **Analyze computational cost for both systems**
e. **Estimate which consumes less power and memory**
f. **Justify which cryptosystem is better for the given scenario with reasoning**

ANS:

Given

RSA =1024-bit key
ECC =160-bit key
Device = IoT

(d) Computational Cost

RSA

- Large key
- Large computations
- Slow on small devices

ECC

- Smaller key
- Faster operations
- Efficient computation

ECC requires significantly fewer mathematical operations.

(e) Power and Memory Consumption

RSA

- Higher RAM usage
- Higher CPU usage
- Higher battery consumption

ECC

- Lower RAM
- Lower CPU
- Lower battery usage

Hence

$$\boxed{\text{ECC consumes less power and memory.}}$$

(f) Recommendation

For an IoT device:

$$\boxed{\text{ECC is the better choice.}}$$

Reasons

- Same security with much smaller keys.
- Faster computation.
- Lower memory usage.
- Lower power consumption.
- Ideal for embedded and battery-powered devices.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation